

数值分析 (MAZ224)

项目报告

NUMERICAL ANALYSIS — PROJECT REPORT

Cover Sheet / 封面

课程名称 Course Name

数值分析 (MAZ224)

组号 Group Number

34

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1	☒	陈永鸿	24344012	chenyh657@mail2.sysu.edu.cn	Signature 签名 <u>陈永鸿</u> Date 日期 <u>2026.6.25</u>
2	☐	边子晔	24344002	bianzy@mail2.sysu.edu.cn	Signature 签名 <u>边子晔</u> Date 日期 <u>2026.6.25</u>
3	☐	黄泓毅	24344034	huanghy296@mail2.sysu.edu.cn	Signature 签名 <u>黄泓毅</u> Date 日期 <u>2026.6.25</u>
4	☐	练泓希	24344062	lianhx3@mail2.sysu.edu.cn	Signature 签名 <u>练泓希</u> Date 日期 <u>2026.6.25</u>

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1 Introduction

This report presents the solutions to the three exercises in the final project of the Numerical Analysis course. Exercise A is about numerical quadrature, exercise B is about iterations for solving linear systems, and exercise C is about solving the two-dimensional Poisson equation. Each exercise includes a description of the problem, the methods used, and the analysis of the results

2 Exercise A

For Exercise A, the target integral is

$$I = \int_L^R f(x) dx, \quad f(x) = ae^{bx} + c \sin(dx) + px^3.$$

The interval $[L, R]$ is divided into n equal subintervals with $h = (R - L)/n$ and nodes $x_i = L + ih$. The code evaluates the integrand vectorially and applies the three required composite rules: the midpoint rule, the trapezoidal rule, and Simpson's rule.

$$M_n = h \sum_{i=0}^{n-1} f\left(\frac{x_i + x_{i+1}}{2}\right),$$
$$T_n = \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right],$$
$$S_n = \frac{h}{3} \left[f(x_0) + 4 \sum_{i \text{ odd}, 1 \leq i \leq n-1} f(x_i) + 2 \sum_{i \text{ even}, 2 \leq i \leq n-2} f(x_i) + f(x_n) \right].$$

The input guarantees that n is even, so Simpson's rule is directly applicable. The public parameters are

$$(L, R, a, b, c, d, p, n) = (-0.5, 1.5, 2.0, -0.5, -2.0, 3.0, 0.5, 120).$$

The public run gives

$$\text{midpoint} = 3.6838831196563744, \quad \text{trapezoid} = 3.684074124645292, \quad \text{simpson} = 3.683946781955994.$$

We also calculate the exact antiderivative

$$F(x) = \frac{a}{b} e^{bx} - \frac{c}{d} \cos(dx) + \frac{p}{4} x^4,$$

and hence the reference value is

$$I = 3.6839467883879182.$$

The errors of the midpoint, trapezoidal, and Simpson results are approximately 6.37×10^{-5} , 1.27×10^{-4} , and 6.43×10^{-9} respectively. This agrees with the expected second-order behavior of midpoint/trapezoidal quadrature and the fourth-order behavior of Simpson quadrature.

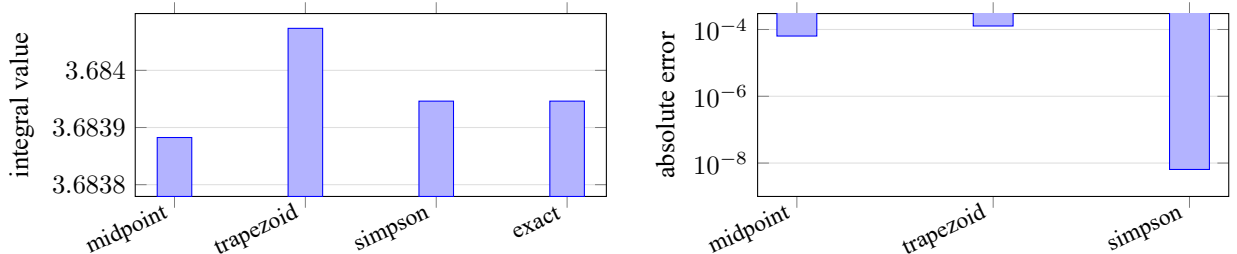


Figure 1: Exercise A numerical quadrature results and absolute errors.

3 Exercise B: Stationary Iterations

Exercise B solves $Ax = b$ for the tridiagonal matrix

$$A = \text{tridiag}(-1, 2 + \mu, -1) \in \mathbb{R}^{n \times n}.$$

The exact solution is:

$$x_j^* = \sin\left(\frac{k\pi(j+1)}{n+1}\right), \quad j = 0, \dots, n-1,$$

which is used to manufacture the right-hand side $b = Ax^*$.

Starting from $x^{(0)} = 0$, the solver applies the method specified by `method`: Jacobi, Gauss-Seidel, or SOR. With $d = 2 + \mu$, the component formulas are

$$x_i^{(m+1)} = \frac{b_i + x_{i-1}^{(m)} + x_{i+1}^{(m)}}{d} \quad (\text{Jacobi}),$$

$$x_i^{(m+1)} = \frac{b_i + x_{i-1}^{(m+1)} + x_{i+1}^{(m)}}{d} \quad (\text{Gauss-Seidel}),$$

$$x_i^{(m+1)} = (1 - \omega)x_i^{(m)} + \omega \frac{b_i + x_{i-1}^{(m+1)} + x_{i+1}^{(m)}}{d} \quad (\text{SOR}).$$

Missing boundary neighbors are treated as zero. After each sweep, we recompute

$$\frac{\|b - Ax^{(m)}\|_2}{\|b\|_2}$$

The iteration stops when this relative residual is at most ε or when `max_iter` is reached.

The public parameters are

$$n = 40, \quad \mu = 0, \quad k = 3, \quad \text{method} = \text{sor}, \quad \omega = 1, \quad \varepsilon = 10^{-6}, \quad \text{max_iter} = 50000.$$

Since $\omega = 1$, the SOR update is exactly Gauss-Seidel in this public run. The computation stops after 958 iterations with

$$\text{residual_norm} = 9.956604319782517 \times 10^{-7},$$

and the first five residuals are

$$[0.9523286626, 0.9061743200, 0.8621977499, 0.8203015725, 0.7803889533].$$

The maximum componentwise difference from x^* is about 8.9×10^{-6} . The full componentwise comparison is shown in Table 1.

Table 1: All components of the final computed solution and exact solution.

j	x_j	x_j^*	abs. err.	j	x_j	x_j^*	abs. err.
0	0.2278527887	0.2278535089	7.20×10^{-7}	20	-0.9934109541	-0.9934020898	8.86×10^{-6}
1	0.4437184060	0.4437198379	1.43×10^{-6}	21	-0.9411488344	-0.9411400480	8.79×10^{-6}
2	0.6362403113	0.6362424423	2.13×10^{-6}	22	-0.8393740837	-0.8393654261	8.66×10^{-6}
3	0.7952900577	0.7952928713	2.81×10^{-6}	23	-0.6934409795	-0.6934325008	8.48×10^{-6}
4	0.9125001408	0.9125036165	3.48×10^{-6}	24	-0.5110269306	-0.5110186794	8.25×10^{-6}
5	0.9817042065	0.9817083200	4.11×10^{-6}	25	-0.3017285752	-0.3017205986	7.98×10^{-6}
6	0.9992614575	0.9992661811	4.72×10^{-6}	26	-0.0765569098	-0.0765492528	7.66×10^{-6}
7	0.9642481932	0.9642534955	5.30×10^{-6}	27	0.1526419899	0.1526492842	7.29×10^{-6}
8	0.8785064044	0.8785122509	5.85×10^{-6}	28	0.3738101808	0.3738170718	6.89×10^{-6}
9	0.7465468682	0.7465532216	6.35×10^{-6}	29	0.5753122105	0.5753186602	6.45×10^{-6}
10	0.5753118402	0.5753186602	6.82×10^{-6}	30	0.7465472484	0.7465532216	5.97×10^{-6}
11	0.3738098279	0.3738170718	7.24×10^{-6}	31	0.8785067864	0.8785122509	5.46×10^{-6}
12	0.1526416613	0.1526492842	7.62×10^{-6}	32	0.9642485687	0.9642534955	4.93×10^{-6}
13	-0.0765572079	-0.0765492528	7.96×10^{-6}	33	0.9992618178	0.9992661811	4.36×10^{-6}
14	-0.3017288372	-0.3017205986	8.24×10^{-6}	34	0.9817045425	0.9817083200	3.78×10^{-6}
15	-0.5110271517	-0.5110186794	8.47×10^{-6}	35	0.9125004434	0.9125036165	3.17×10^{-6}
16	-0.6934411556	-0.6934325008	8.65×10^{-6}	36	0.7952903177	0.7952928713	2.55×10^{-6}
17	-0.8393742118	-0.8393654261	8.79×10^{-6}	37	0.6362405196	0.6362424423	1.92×10^{-6}
18	-0.9411489122	-0.9411400480	8.86×10^{-6}	38	0.4437185535	0.4437198379	1.28×10^{-6}
19	-0.9934109802	-0.9934020898	8.89×10^{-6}	39	0.2278528667	0.2278535089	6.42×10^{-7}

Optimal SOR parameter analysis.

The public input uses $\omega = 1$, so SOR reduces to Gauss-Seidel. It is still useful to analyze the optimal relaxation parameter, because SOR can be much faster for the same tridiagonal Toeplitz matrix. Writing

$$A = D - L - U,$$

the Jacobi iteration matrix is

$$B_J = D^{-1}(L + U).$$

For

$$A = \text{tridiag}(-1, 2 + \mu, -1),$$

the eigenvalues of B_J are

$$\lambda_j(B_J) = \frac{2}{2 + \mu} \cos\left(\frac{j\pi}{n+1}\right), \quad j = 1, \dots, n.$$

Therefore the spectral radius is

$$\rho_J = \frac{2}{2 + \mu} \cos\left(\frac{\pi}{n+1}\right).$$

When A is consistently ordered and the Jacobi spectral radius satisfies $\rho_J < 1$, the classical optimal SOR parameter is

$$\omega_* = \frac{2}{1 + \sqrt{1 - \rho_J^2}}.$$

For the public parameters $n = 40$ and $\mu = 0$, this gives

$$\rho_J = \cos\left(\frac{\pi}{41}\right) = 0.9970658012,$$

$$\omega_* = \frac{2}{1 + \sqrt{1 - \rho_J^2}} = 1.8577877368.$$

Using the same stopping tolerance, the observed comparison is shown in Table 2.

Table 2: Comparison of stationary iterations for Exercise B.

method	ω	iterations	final relative residual
Jacobi	—	519	9.8085×10^{-7}
Gauss-Seidel	—	958	9.9566×10^{-7}
SOR	1.0000	958	9.9566×10^{-7}
SOR with ω_*	1.8578	108	9.9248×10^{-7}

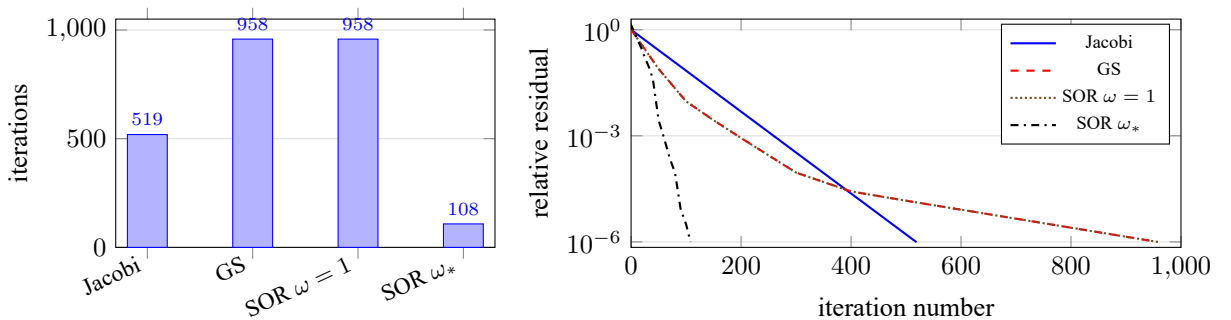


Figure 2: Iteration counts and residual histories for the four Exercise B runs.

The optimized SOR method is the most efficient in this comparison. The Jacobi method gives a smaller error in this specific public test because the exact solution is precisely an eigenvector of matrix A , which Jacobi preserves exactly as a single mode.

4 Exercise C: Two-Dimensional Poisson Equation

Exercise C solves

$$-\Delta u = f \quad \text{in } (0, 1)^2, \quad u = 0 \quad \text{on } \partial(0, 1)^2,$$

Using exact solution

$$u_{\text{exact}}(x, y) = \sin(m\pi x) \sin(n\pi y).$$

We manufacture the right-hand side

$$f(x, y) = \pi^2(m^2 + n^2) \sin(m\pi x) \sin(n\pi y).$$

On an $N \times N$ interior grid with $h = 1/(N + 1)$, the five-point finite-difference system is

$$4u_{i,j} - u_{i-1,j} - u_{i+1,j} - u_{i,j-1} - u_{i,j+1} = h^2 f(x_i, y_j).$$

The right-hand side of the discrete system is then flattened into a vector

$$b = h^2 f.$$

We map the grid point (i, j) to the vector index $iN + j$ and explicitly forms the corresponding matrix A .

$$A = \begin{bmatrix} T & -I & 0 & \cdots & 0 \\ -I & T & -I & \cdots & 0 \\ 0 & -I & T & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & T \end{bmatrix} \quad T = \text{tridiag}(-1, 4, -1) \in \mathbb{R}^{N \times N}$$

Boundary values do not appear as unknowns because the boundary condition is homogeneous, $u = 0$.

The linear system $Au = b$ is solved by conjugate-gradient method. Starting from $u^{(0)} = 0$, it uses

$$\begin{aligned} r^{(0)} &= b - Au^{(0)}, & p^{(0)} &= r^{(0)}, \\ \alpha_k &= \frac{(r^{(k)})^T r^{(k)}}{(p^{(k)})^T A p^{(k)}}, & u^{(k+1)} &= u^{(k)} + \alpha_k p^{(k)}, \\ r^{(k+1)} &= r^{(k)} - \alpha_k A p^{(k)}, & \beta_k &= \frac{(r^{(k+1)})^T r^{(k+1)}}{(r^{(k)})^T r^{(k)}}, & p^{(k+1)} &= r^{(k+1)} + \beta_k p^{(k)}. \end{aligned}$$

The public parameters are $N = 9$, $m = 3$, and $n = 3$. The public run gives

```
center_value      1.0774291810736978
max_error         0.0774291810736978
residual_inf      2.6645352591003757 × 10-15.
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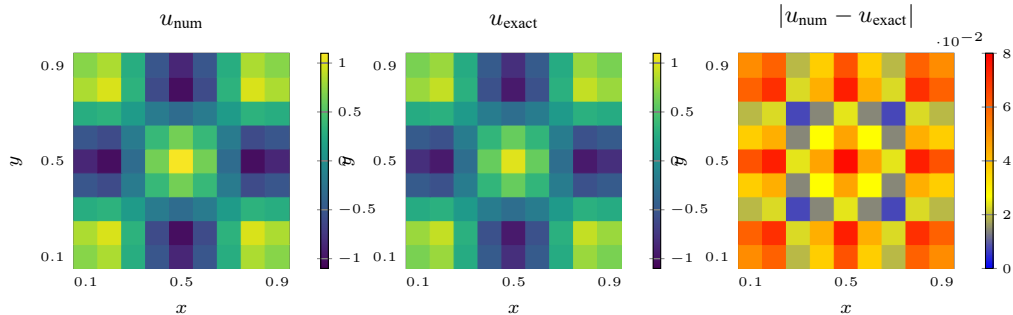


Figure 3: Heat maps of the Exercise C numerical solution, exact solution, and absolute error on the 9×9 interior grid.

One-step CG convergence. The one-step convergence in the public case is not accidental. The two-dimensional five-point matrix has discrete sine eigenvectors

$$v_{p,q}(i,j) = \sin\left(\frac{p\pi(i+1)}{N+1}\right) \sin\left(\frac{q\pi(j+1)}{N+1}\right),$$

for $i, j = 0, \dots, N-1$. The corresponding eigenvalues are

$$\lambda_{p,q} = 4 - 2\cos\left(\frac{p\pi}{N+1}\right) - 2\cos\left(\frac{q\pi}{N+1}\right).$$

For the public parameters, the sampled exact solution is exactly the single eigenvector $v_{3,3}$. Since

$$b = h^2\pi^2(m^2 + n^2)v_{m,n},$$

the right-hand side is also a scalar multiple of the same eigenvector. With the zero initial guess,

$$r^{(0)} = b, \quad p^{(0)} = b.$$

Because $Ab = \lambda_{m,n}b$, the first CG step has

$$\alpha_0 = \frac{b^T b}{b^T Ab} = \frac{1}{\lambda_{m,n}}.$$

Therefore

$$u^{(1)} = u^{(0)} + \alpha_0 p^{(0)} = \frac{b}{\lambda_{m,n}}, \quad Au^{(1)} = b.$$

The residual is consequently zero up to floating-point roundoff after one iteration. This explains why the reported infinity-norm residual is about 10^{-15} .

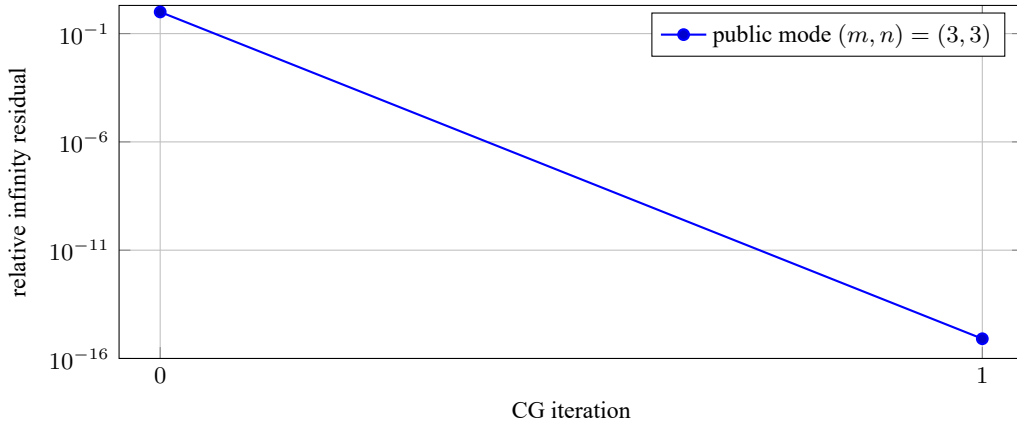


Figure 4: Relative residual history for the public Exercise C right-hand side. The residual reaches roundoff after one CG step because the right-hand side is a single discrete sine eigenmode.

5 Conclusion

The three exercises satisfy the requested numerical tasks. Exercise A shows the expected accuracy hierarchy of composite quadrature rules. Exercise B satisfies the residual stopping criterion for the public SOR run, and the optimal-parameter analysis explains how SOR can be accelerated from 958 iterations to about 108 iterations for the same tolerance. Exercise C solves the five-point Poisson system to roundoff residual; its one-step CG convergence follows from the discrete sine eigenstructure of the public manufactured solution.